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Place your plot of the PD controller's response to a static setpoint.

Add a brief caption (2-3 sentences) that includes the K_p and K_d constants you used, what critical damping is, and any difficulties you had with tuning the controller

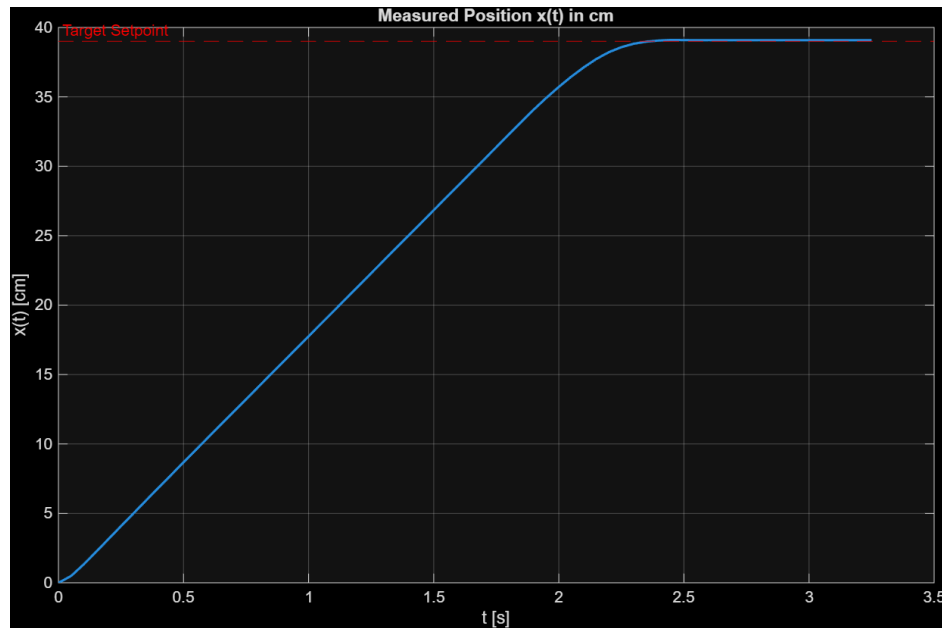


Figure 1: this is the $x(t)$ plot with a 39cm setpoint.

This was pretty difficult to get done at first since I was trying to do too much with the values I chose. I restarted and got where I needed to be.

$$k_p = 9$$

$$k_d = 1$$

$$\zeta \approx 1.0$$

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Figure 2: dynamic setpoint plot

This is the response and dynamic setpoint. I did need to change them from the static setpoint values from before. k_p seemed to be how "hard" the motor was able to turn back around. Maybe using the word sharpness would make more sense. k_d was the damping that went along with that acceleration.

$$k_p = 13.48$$

$$k_i = 0$$

$$k_d = 2$$

- What does changing K_p do to the system response? What about K_i ? K_d ? k_d would change how much "far" the position reached. k_i would change how much SS error there was which I didnt really use. k_d would change how the quickly I could stabilize
- For your dynamic setpoint, how did the system respond when you added K_i ? How did you tune K_p and K_d in response to this change? The system didnt really change from k_i at least not meaningfully for me, I just treated k_p and k_d like normal and slowly adjusted
- In a system where the setpoint is consistently larger or smaller than the control input, why would implementing K_i lead to overshoot (Hint: this is called integral windup)? If we were coding a PID controller from scratch, how could we account for this I feel like this could be fixed by hard coding the error percentage analysis into the loop. This way

we can see that we are off by an error percentage and then we can proportionally increase or decrease the power applied.

Page 3:

List online sources and collaborators.

<https://playground.arduino.cc/Code/PIDLibrary>

PonMExample/

<https://www.teachmemicro.com/arduino-pid-control-tutorial/>

<https://www.mathworks.com/matlabcentral/answers/418825-damping-factor-and-natural-frequency-out-of-time-response-data>

I had chatGPT help with the normalizing of the data and plots. To make the measuring easier.

I had no help from other students.

Page 4: Self-evaluation:

- (a) What did you expect to learn? I was really hoping to be able to understand how this would tie into the transfer function material from other classes. It was more realistic and not what I expected.
- (b) How would you rate your attendance and participation in class during this lab experience? Are any adjustments needed? I did great this week, thankfully nothing crazy was going on.
- (c) How well did you do at the lab assignment itself? You might include thinking about strengths and weaknesses of your performance in this response. Please also give yourself a grade (out of 4 possible points). I would give myself a 4 for this. I kept on top of everything.
- (d) What did you actually learn in the lab? (This might include a range of things, from subject matter and concrete skills to ways of knowing and working together with peers.) I got a better appreciation for trouble shooting. I had a lot more occurrences with the arduino not acting right for this lab. I also got to see how changing the k params would alter the response.

Code

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